Capstone Project |E-commerce_Customer_Segmentation_and_Prediction

This project focuses on analyzing e-commerce customer transaction data to segment customers based on their purchasing behavior and predict high-value customers using machine learning techniques.

Introduction:

Problem:

Project Workflow

- Step 1 | Import Libraries
- Step 2 | Read Dataset
- Step 3 | Dataset Overview
- Step 4 | Data Preprocessing
- Step 5 | Exploratory Data Analysis
- Step 6 | Feature Scaling
- Step 7 | Random Forest
- Step 8 | Logistic Regression
- Step 9 | Model Comparison
- Step 10 | Conclusion

Dataset Description

The dataset contains transactional data from an e-commerce platform. Each record represents a customer purchase and includes invoice details, product information, quantity, transaction date, unit price, customer ID, and country.

Let's get started:

Importing Required Libraries

Step 1 | Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import os

%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.colors import ListedColormap

from sklearn.model_selection import train_test_split
from scipy.stats import boxcox
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler,OneHotEncoder
from sklearn.model_selection import GridSearchCV, StratifiedKFold

from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import LabelEncoder



from sklearn.metrics import classification_report, accuracy_score, confusion_

import ipywidgets as widgets
from IPython.display import display, HTML
import joblib


import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
os.getcwd()
```

Out[2]:
```
'C:\\Users\\ajoyb'
```

In [3]:
```
os.chdir(r"C:\BLA Capston Projects")
```

In [4]:
```
os.getcwd()
```

Out[4]:
```
'C:\\BLA Capston Projects'
```

Step 2 | Read Dataset

load The CSV Dataset

In [5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [6]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

Data Loading

In [7]:
```
df = pd.read_csv(
    "capston project data.csv",
    encoding="latin1"
)
```

Dataset Description

Data Summary & Description

The dataset used in this project consists of transactional records from an e-commerce platform. Each record represents a single purchase made by a customer. The dataset is used to analyze customer purchasing behavior, perform customer segmentation, and predict high-value customers.

Descriptive Statistics for Numerical Features

Feature	Description
Quantity	Number of products purchased in a single transaction
UnitPrice	Price of a single unit of a product
TotalAmount	Total transaction value (Quantity × UnitPrice)
Recency	Number of days since the customer's last purchase
Frequency	Total number of transactions made by the customer
Monetary	Total amount spent by the customer

Variable Descriptions (E-commerce Dataset)

Variable	Description
InvoiceNo	Unique identifier assigned to each purchase transaction
StockCode	Unique code assigned to each product
Description	Name or description of the purchased product

Quantity	Number of units purchased in the transaction
InvoiceDate	Date and time when the transaction occurred
UnitPrice	Price per unit of the product
CustomerID	Unique identifier assigned to each customer
Country	Country from which the customer made the purchase
HighValueCustomer	Target Variable indicating customer value 1 = High-Value Customer 0 = Low-Value Customer

Step 3 | Dataset Overview

Data Understanding

Now A Deep Dive into Transaction Patterns:

Step 3 | Dataset Basic Information

In [8]:
```
os.listdir()
```

Out[8]:
```
['capston project data.csv',
 'description1.txt',
 'gradient_boosting_model.pkl',
 'logistic_regression_model.pkl',
 'random_forest_model.pkl']
```

In [9]:
```
df
```

Out[9]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	12/9/2011 12:50	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	12/9/2011 12:50	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	12/9/2011 12:50	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	12/9/2011 12:50	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	12/9/2011 12:50	4.95	1

541909 rows × 8 columns

Step 3 | Dataset Overview

Now A Deep Dive into Transaction Patterns:

Step 3 | Dataset Basic Information

Display a concise summary of the dataframe

In [10]:
```
df.head()
```

Out[10]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	12/1/2010 8:26	3.39	17850.0
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	12/1/2010 8:26	3.39	17850.0

In [11]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  object
 1   StockCode    541909 non-null  object
 2   Description  540455 non-null  object
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  object
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   406829 non-null  float64
 7   Country      541909 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

In [12]:
```
df.describe()
```

Out[12]:

	Quantity	UnitPrice	CustomerID
count	541909.000000	541909.000000	406829.000000
mean	9.552250	4.611114	15287.690570
std	218.081158	96.759853	1713.600303
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13953.000000
50%	3.000000	2.080000	15152.000000
75%	10.000000	4.130000	16791.000000
max	80995.000000	38970.000000	18287.000000

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
InvoiceNo           0
StockCode           0
Description      1454
Quantity            0
InvoiceDate         0
UnitPrice           0
CustomerID     135080
Country             0
dtype: int64
```

Inferences from Data Info

- **Number of Entries:** The dataset contains a large number of transactional records, where each row represents a single purchase made by a customer on the e-commerce platform.

- **Columns:** The dataset includes multiple columns related to transaction details, product information, customer identifiers, pricing, and geographic location.

- **Data Types:**
  - **Numerical Columns:** Columns such as *Quantity*, *UnitPrice*, and *TotalAmount* are numerical and are suitable for statistical analysis, scaling, and modeling.
  - **Categorical Columns:** Columns like *InvoiceNo*, *StockCode*, *Description*, and *Country* are categorical in nature and require encoding or aggregation before modeling.

- **Missing Values:** After initial inspection, missing values were handled appropriately to ensure data consistency. This step ensures the dataset is clean and ready for feature engineering and modeling.

- **Data Readiness:** The dataset is well-structured and suitable for further analysis such as exploratory data analysis, RFM feature engineering, customer segmentation, and predictive modeling.

Step 4 | Data Preprocessing

Data Preprocessing

In [14]:
```
df = df.dropna(subset=['CustomerID'])
df = df[df['Quantity'] > 0]
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
df['TotalAmount'] = df['Quantity'] * df['UnitPrice']
```

In [15]:
```
df
```

Out[15]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 9 columns

In [16]:
```
df.shape
```

Out[16]:
```
(397924, 9)
```

In [17]:
```
print(df.head().to_markdown(index=False, numalign="left", stralign="left"))
print(df.info())
```

```
| InvoiceNo   | StockCode   | Description                         | Quantity
| InvoiceDate         | UnitPrice   | CustomerID   | Country        | TotalAmo
unt   |
|:------------|:------------|:------------------------------------|:----------
-|:--------------------|:------------|:-------------|:---------------|:-------
-------|
| 536365      | 85123A      | WHITE HANGING HEART T-LIGHT HOLDER  | 6
| 2010-12-01 08:26:00 | 2.55        | 17850        | United Kingdom | 15.3
|
| 536365      | 71053       | WHITE METAL LANTERN                 | 6
| 2010-12-01 08:26:00 | 3.39        | 17850        | United Kingdom | 20.34
|
| 536365      | 84406B      | CREAM CUPID HEARTS COAT HANGER      | 8
| 2010-12-01 08:26:00 | 2.75        | 17850        | United Kingdom | 22
|
| 536365      | 84029G      | KNITTED UNION FLAG HOT WATER BOTTLE | 6
| 2010-12-01 08:26:00 | 3.39        | 17850        | United Kingdom | 20.34
|
| 536365      | 84029E      | RED WOOLLY HOTTIE WHITE HEART.      | 6
| 2010-12-01 08:26:00 | 3.39        | 17850        | United Kingdom | 20.34
|
<class 'pandas.core.frame.DataFrame'>
Index: 397924 entries, 0 to 541908
Data columns (total 9 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    397924 non-null  object
 1   StockCode    397924 non-null  object
 2   Description  397924 non-null  object
 3   Quantity     397924 non-null  int64
 4   InvoiceDate  397924 non-null  datetime64[ns]
 5   UnitPrice    397924 non-null  float64
 6   CustomerID   397924 non-null  float64
 7   Country      397924 non-null  object
 8   TotalAmount  397924 non-null  float64
dtypes: datetime64[ns](1), float64(3), int64(1), object(4)
memory usage: 30.4+ MB
None
```

Categorical Data Analysis

In [18]:
```
print(df.columns)
```

```
Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'UnitPrice', 'CustomerID', 'Country', 'TotalAmount'],
      dtype='object')
```

In [19]:
```
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']
```

In [20]:
```
df
```

Out[20]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

In [21]:
```
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

In [22]:
```
df
```

Out[22]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

In [23]:

```
reference_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)

rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (reference_date - x.max()).days,
    'InvoiceNo': 'nunique',
```

```
    'TotalPrice': 'sum'
})
rfm.columns = ['Recency', 'Frequency', 'Monetary']
```

In [24]:
```
rfm
```

Out[24]:

	Recency	Frequency	Monetary
CustomerID			
12346.0	326	1	77183.60
12347.0	2	7	4310.00
12348.0	75	4	1797.24
12349.0	19	1	1757.55
12350.0	310	1	334.40
...	...	...	...
18280.0	278	1	180.60
18281.0	181	1	80.82
18282.0	8	2	178.05
18283.0	4	16	2094.88
18287.0	43	3	1837.28

4339 rows × 3 columns

In [25]:
```
monetary_threshold = rfm['Monetary'].quantile(0.75)
rfm['HighValueCustomer'] = (rfm['Monetary'] >= monetary_threshold).astype(int
```

In [26]:
```
monetary_threshold
```

Out[26]:
```
np.float64(1661.64)
```

In [27]:
```
import pandas as pd

# Make sure df exists
# df = pd.read_csv("your_dataset.csv")  # only if not already loaded

# Convert InvoiceDate to datetime
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])

# Reference date
reference_date = df['InvoiceDate'].max() + pd.Timedelta(days=1)

# Create RFM table
rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (reference_date - x.max()).days,
    'InvoiceNo': 'nunique',
    'TotalPrice': 'sum'
})

rfm.columns = ['Recency', 'Frequency', 'Monetary']
```

In [28]:
```
rfm.columns
```

Out[28]:
```
Index(['Recency', 'Frequency', 'Monetary'], dtype='object')
```

In [29]:
```
# Define High-Value customers using Monetary value (Top 25%)
monetary_threshold = rfm['Monetary'].quantile(0.75)
rfm['HighValueCustomer'] = (rfm['Monetary'] >= monetary_threshold).astype(int
```

In [30]:
```
rfm['HighValueCustomer']
```

Out[30]:
```
CustomerID
12346.0    1
12347.0    1
12348.0    1
12349.0    1
12350.0    0
          ..
18280.0    0
18281.0    0
18282.0    0
18283.0    1
18287.0    1
Name: HighValueCustomer, Length: 4339, dtype: int64
```

In [31]:
```
# 1. Analyze Customer Value Imbalance
high_value_percentage = round(
    (rfm["HighValueCustomer"].sum() / rfm.shape[0]) * 100, 2
)

# 2. Country-wise Customer Distribution (UNIQUE customers)
country_counts = df.groupby('Country')['CustomerID'].nunique().reset_index()
country_counts.columns = ['Country', 'total_customers']

# High-value customer count per country
high_value_by_country = df.merge(
    rfm[['HighValueCustomer']],
    left_on='CustomerID',
    right_index=True,
    how='inner'
).groupby('Country')['HighValueCustomer'].sum().reset_index()

high_value_by_country.columns = ['Country', 'high_value_customers']

# Merge results
country_analysis = pd.merge(
    country_counts,
    high_value_by_country,
    on='Country',
    how='left'
)

# Calculate percentage of high-value customers
country_analysis['high_value_percentage'] = round(
    (country_analysis['high_value_customers'] / country_analysis['total_custo
)

# Print results
print("\n--- Customer Value Distribution Analysis ---")
print(f"Total Customers: {rfm.shape[0]}")
```

```
print(f"High-Value Customers: {rfm['HighValueCustomer'].sum()}")
print(f"Percentage of High-Value Customers: {high_value_percentage}%")

print("\n--- Country-wise High-Value Customer Analysis ---")
print(country_analysis.head(10).to_markdown(index=False, numalign='left', str
```

```
--- Customer Value Distribution Analysis ---
Total Customers: 4339
High-Value Customers: 1085
Percentage of High-Value Customers: 25.01%

--- Country-wise High-Value Customer Analysis ---
| Country         | total_customers   | high_value_customers   | high_value_pe
rcentage   |
|:----------------|:------------------|:-----------------------|:-------------
-----------|
| Australia       | 9                 | 1028                   | 11422.2
|
| Austria         | 11                | 158                    | 1436.36
|
| Bahrain         | 2                 | 0                      | 0
|
| Belgium         | 25                | 1557                   | 6228
|
| Brazil          | 1                 | 0                      | 0
|
| Canada          | 4                 | 135                    | 3375
|
| Channel Islands | 9                 | 589                    | 6544.44
|
| Cyprus          | 8                 | 451                    | 5637.5
|
| Czech Republic  | 1                 | 0                      | 0
|
| Denmark         | 9                 | 217                    | 2411.11
|
```

In [32]:

```
for col in df.select_dtypes(include=["object","category"]).columns:
    print("Column:", col)
    print("Unique Values:", df[col].nunique())
    print()
```

```
Column: InvoiceNo
Unique Values: 18536

Column: StockCode
Unique Values: 3665

Column: Description
Unique Values: 3877

Column: Country
Unique Values: 37
```

FINDING DISTINCT CATEGORIES IN THE CATEGORICAL

In [33]:
```
df["Country"].unique()
```

Out[33]:
```
array(['United Kingdom', 'France', 'Australia', 'Netherlands', 'Germany',
       'Norway', 'EIRE', 'Switzerland', 'Spain', 'Poland', 'Portugal',
       'Italy', 'Belgium', 'Lithuania', 'Japan', 'Iceland',
       'Channel Islands', 'Denmark', 'Cyprus', 'Sweden', 'Finland',
       'Austria', 'Greece', 'Singapore', 'Lebanon',
       'United Arab Emirates', 'Israel', 'Saudi Arabia', 'Czech Republic',
       'Canada', 'Unspecified', 'Brazil', 'USA', 'European Community',
       'Bahrain', 'Malta', 'RSA'], dtype=object)
```

In [34]:
```
df["Description"].nunique()
```

Out[34]:
```
3877
```

In [35]:
```
df["StockCode"].nunique()
```

Out[35]:
```
3665
```

In [36]:
```
df["InvoiceNo"].nunique()
```

Out[36]:
```
18536
```

In [37]:
```
df.dtypes
```

Out[37]:
```
InvoiceNo              object
StockCode              object
Description            object
Quantity                int64
InvoiceDate    datetime64[ns]
UnitPrice             float64
CustomerID            float64
Country                object
TotalAmount           float64
TotalPrice            float64
dtype: object
```

In [38]:
```
df.columns
```

Out[38]:
```
Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'UnitPrice', 'CustomerID', 'Country', 'TotalAmount', 'TotalPrice'],
      dtype='object')
```

Finding distinct categories in categorical columns

In [39]:
```
for col in df.select_dtypes(include=["object", "category"]).columns:
    print("Column Name:", col)
    print("Number of Unique Values:", df[col].nunique())
    print("Sample Categories:", df[col].dropna().unique()[:5])
    print("----------------------------------------")
```

```
Column Name: InvoiceNo
Number of Unique Values: 18536
Sample Categories: ['536365' '536366' '536367' '536368' '536369']
----------------------------------------
Column Name: StockCode
Number of Unique Values: 3665
Sample Categories: ['85123A' '71053' '84406B' '84029G' '84029E']
----------------------------------------
Column Name: Description
Number of Unique Values: 3877
Sample Categories: ['WHITE HANGING HEART T-LIGHT HOLDER' 'WHITE METAL LANTERN'
 'CREAM CUPID HEARTS COAT HANGER' 'KNITTED UNION FLAG HOT WATER BOTTLE'
 'RED WOOLLY HOTTIE WHITE HEART.']
----------------------------------------
Column Name: Country
Number of Unique Values: 37
Sample Categories: ['United Kingdom' 'France' 'Australia' 'Netherlands' 'Germa
ny']
----------------------------------------
```

Data Cleaning

Remove missing customers

In [40]:
```
df = df.dropna(subset=['CustomerID'])
```

In [41]:
```
df
```

Out[41]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

Remove cancelled orders

In [42]:
```
df = df[df['Quantity'] > 0]
```

In [43]:
```
df
```

Out[43]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

Convert date

In [44]:
```
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

In [45]:
```
df
```

Out[45]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

Create total amount

In [46]:
```
df['TotalAmount'] = df['Quantity'] * df['UnitPrice']
```

In [47]:
```
df
```

Out[47]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 10 columns

Step 5 | Exploratory Data Analysis(EDA)

Inferences from Data Info

1. **Data Quality and Reliability:** The dataset is well-structured and contains complete transactional records after preprocessing. Missing values and invalid entries were handled appropriately, providing a reliable foundation for customer behavior analysis and predictive modeling.

2. **Customer Value Imbalance:** The dataset exhibits an imbalance between high-value and low-value customers, where only a smaller proportion of customers contribute significantly to overall revenue. This imbalance is important for modeling, as it highlights the need to focus on identifying and retaining high-value customers rather than treating all customers equally.

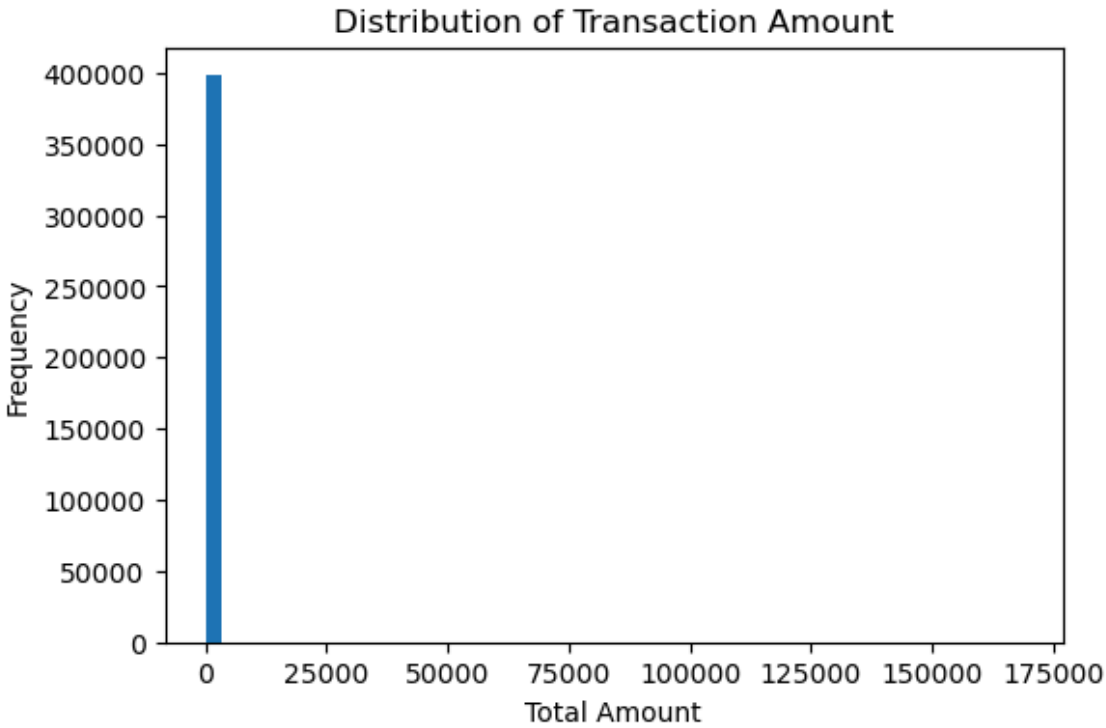
3. **Business-Oriented Insights:** Analysis of purchasing behavior shows that customer value is primarily driven by **purchase frequency** and **monetary value**. These insights enable the business to:
   - Identify and reward loyal, high-spending customers.
   - Design targeted marketing campaigns for medium-value customers.
   - Implement re-engagement strategies for low-frequency or inactive customers.

Exploratory Data Analysis

In [48]:
```
df.columns
```

Out[48]:
```
Index(['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'UnitPrice', 'CustomerID', 'Country', 'TotalAmount', 'TotalPrice'],
      dtype='object')
```

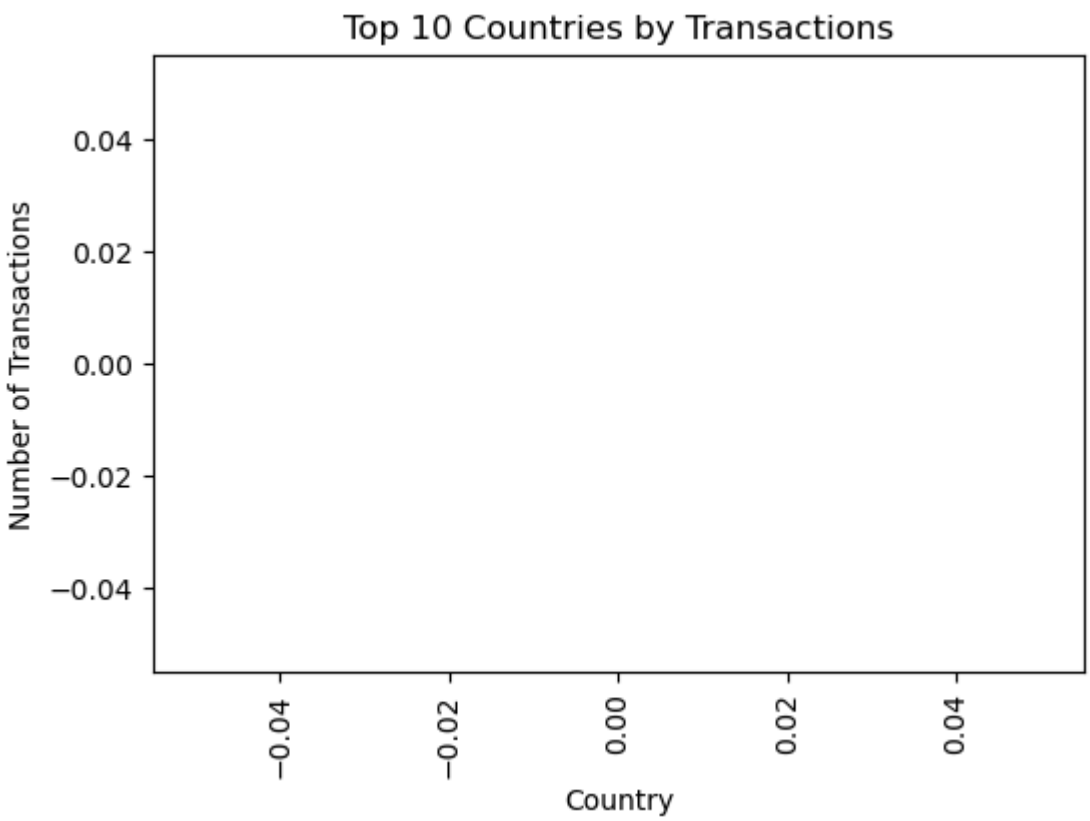
In [49]:
```
plt.figure(figsize=(6,4))
plt.hist(df["TotalAmount"], bins=50)
plt.xlabel("Total Amount")
plt.ylabel("Frequency")
plt.title("Distribution of Transaction Amount")
plt.show()
```



In [50]:
```
country_cols = [col for col in df.columns if col.startswith("Country_")]
print(len(country_cols))
```

```
0
```

In [51]:
```
import matplotlib.pyplot as plt
country_cols = [col for col in df.columns if col.startswith("Country_")]
top_countries = (
    df[country_cols]
    .sum()
    .sort_values(ascending=False)
    .head(10)
)
plt.figure(figsize=(6,4))
plt.bar(top_countries.index, top_countries.values)
plt.xlabel("Country")
plt.ylabel("Number of Transactions")
plt.title("Top 10 Countries by Transactions")
plt.xticks(rotation=90)
plt.show()
```



In [52]:
```
rfm['HighValueCustomer'].value_counts().plot(
    kind='bar',
    title='High-Value vs Low-Value Customers'
)
```

Out[52]:
```
<Axes: title={'center': 'High-Value vs Low-Value Customers'}, xlabel='HighVa
lueCustomer'>
```

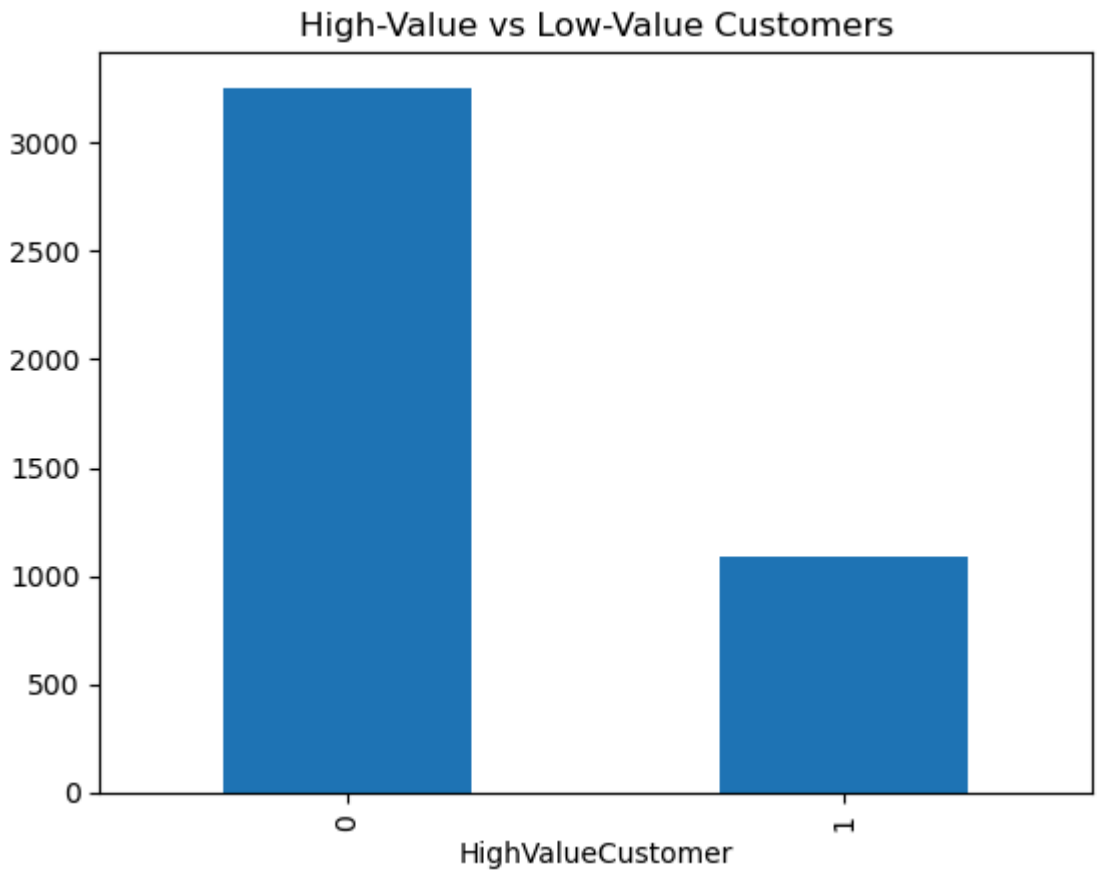
In [53]:
```
# Total customers
total_customers = rfm.shape[0]

# High-value customers
high_value_customers = rfm['HighValueCustomer'].sum()

# Percentage of high-value customers
high_value_percent = round(
    (high_value_customers / total_customers) * 100, 2
)

print("Total Customers:", total_customers)
print("High-Value Customers:", high_value_customers)
print("Percentage of High-Value Customers:", high_value_percent, "%")
```

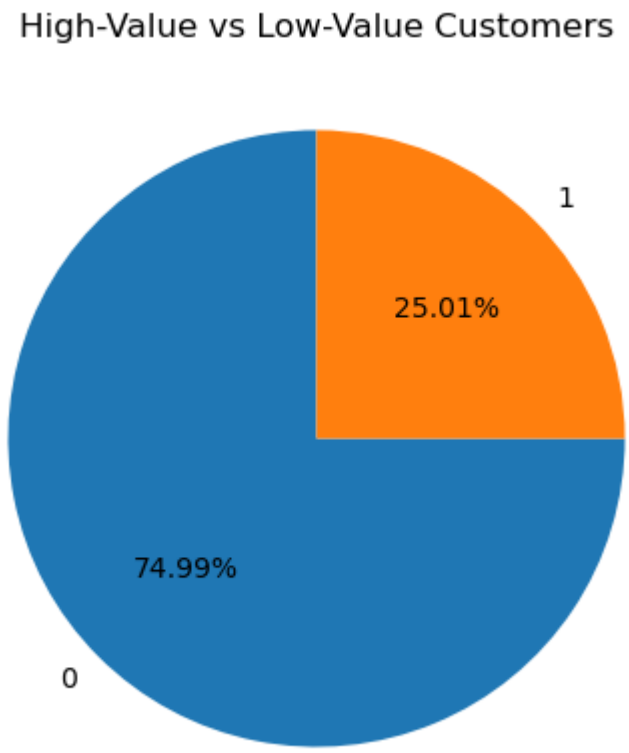
```
Total Customers: 4339
High-Value Customers: 1085
Percentage of High-Value Customers: 25.01 %
```

Pie chart

In [54]:
```
plt.figure(figsize=(5,5))
rfm['HighValueCustomer'].value_counts().plot(
    kind='pie',
    autopct='%1.2f%%',
```

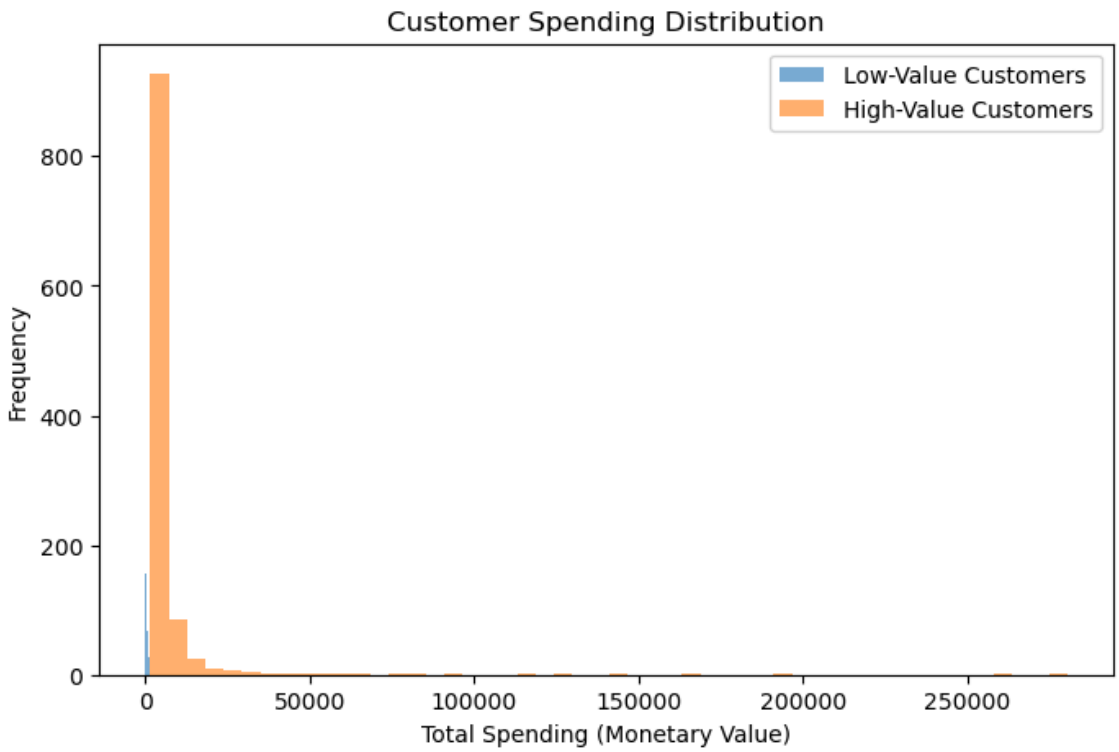
```
    startangle=90
)
plt.title("High-Value vs Low-Value Customers")
plt.ylabel("")
plt.show()
```





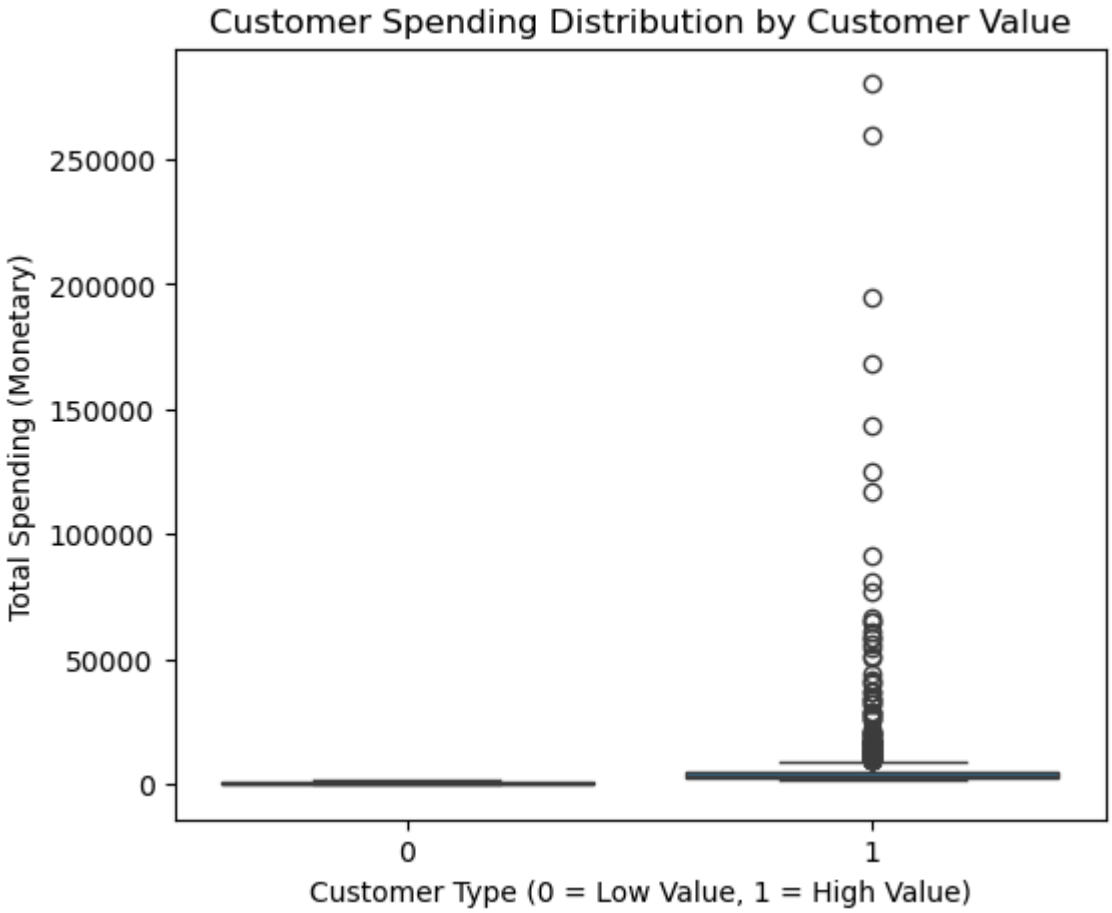
Analyzing

In [55]:
```
plt.figure(figsize=(8,5))
plt.hist(
    rfm[rfm['HighValueCustomer'] == 0]['Monetary'],
    bins=50,
    alpha=0.6,
    label='Low-Value Customers'
)
plt.hist(
    rfm[rfm['HighValueCustomer'] == 1]['Monetary'],
    bins=50,
    alpha=0.6,
    label='High-Value Customers'
)
plt.legend()
plt.title("Customer Spending Distribution")
plt.xlabel("Total Spending (Monetary Value)")
plt.ylabel("Frequency")
plt.show()
```



Boxplot of Transaction Amounts

In [56]:
```
plt.figure(figsize=(6,5))
sns.boxplot(
    x='HighValueCustomer',
    y='Monetary',
    data=rfm
)
plt.title("Customer Spending Distribution by Customer Value")
plt.xlabel("Customer Type (0 = Low Value, 1 = High Value)")
plt.ylabel("Total Spending (Monetary)")
plt.show()
```



E-COMMERCE MAPPING

In [57]:
```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

In [58]:
```
print(rfm.head())
print(rfm.columns)
```

```
            Recency  Frequency  Monetary  HighValueCustomer
CustomerID
12346.0         326          1  77183.60                  1
12347.0           2          7   4310.00                  1
12348.0          75          4   1797.24                  1
12349.0          19          1   1757.55                  1
12350.0         310          1    334.40                  0
Index(['Recency', 'Frequency', 'Monetary', 'HighValueCustomer'], dtype='objec
t')
```

In [59]:
```
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(
    rfm[['Recency', 'Frequency', 'Monetary']]
)
```

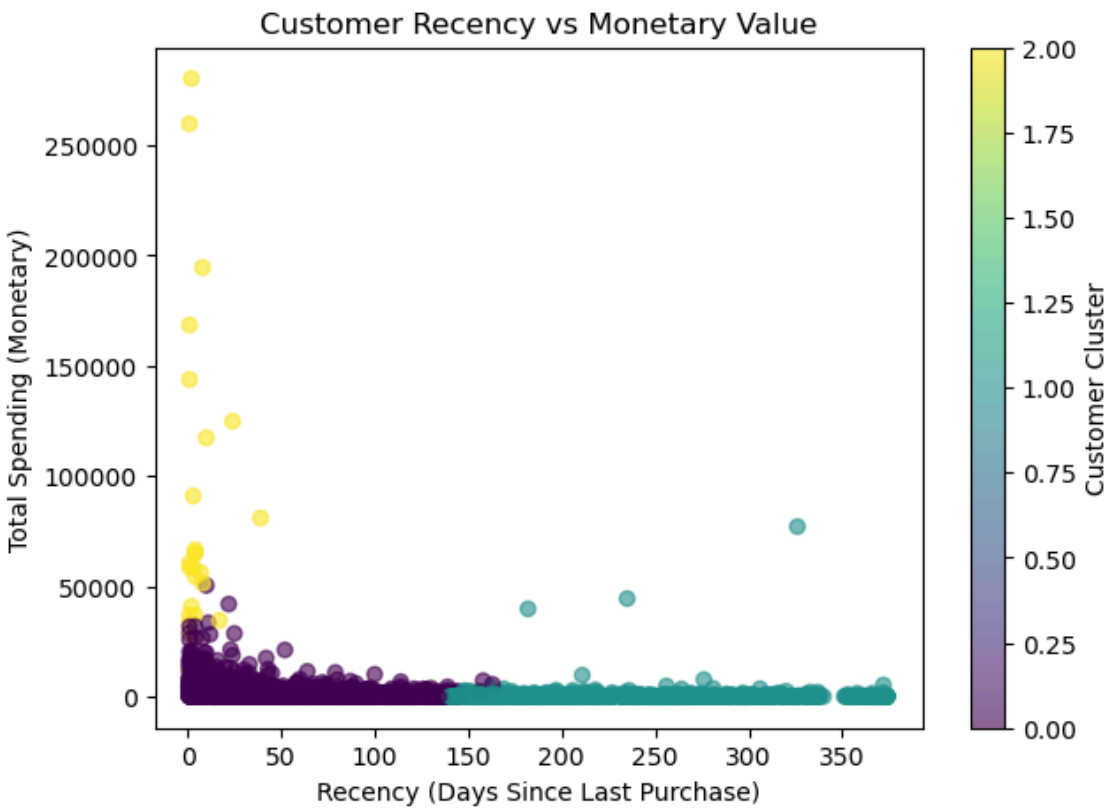
In [60]:
```
kmeans = KMeans(n_clusters=3, random_state=42)
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

```
  File "C:\Users\ajoyb\anaconda3\Lib\site-packages\joblib\externals\loky\backe
nd\context.py", line 257, in _count_physical_cores
    cpu_info = subprocess.run(
        "wmic CPU Get NumberOfCores /Format:csv".split(),
        capture_output=True,
        text=True,
    )
  File "C:\Users\ajoyb\anaconda3\Lib\subprocess.py", line 554, in run
    with Popen(*popenargs, **kwargs) as process:
         ~~~~~^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ajoyb\anaconda3\Lib\subprocess.py", line 1039, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
    ~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
                        pass_fds, cwd, env,
                        ^^^^^^^^^^^^^^^^^^^
    ...<5 lines>...
                        gid, gids, uid, umask,
                        ^^^^^^^^^^^^^^^^^^^^^^
                        start_new_session, process_group)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\Users\ajoyb\anaconda3\Lib\subprocess.py", line 1554, in _execute_ch
ild
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
                       ~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
                             # no special security
                             ^^^^^^^^^^^^^^^^^^^^^
    ...<4 lines>...
                             cwd,
                             ^^^^
                             startupinfo)
                             ^^^^^^^^^^^^
```

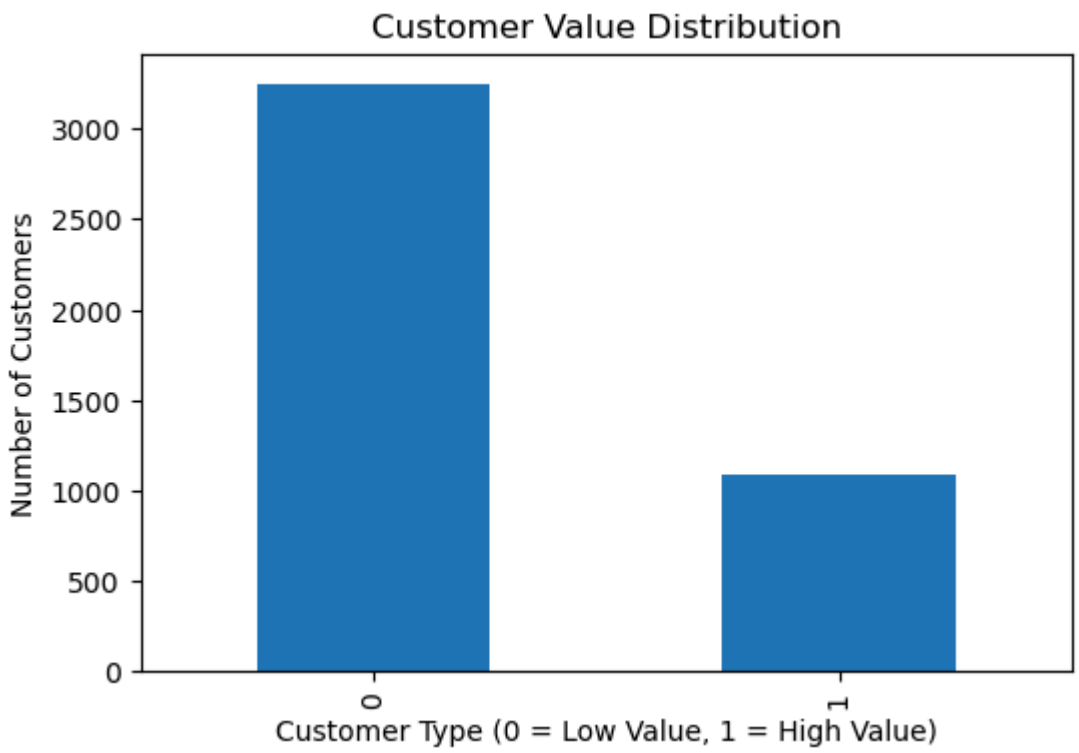
In [61]:
```
print(rfm[['Recency', 'Monetary', 'Cluster']].head())
```

```
            Recency  Monetary  Cluster
CustomerID
12346.0         326  77183.60        1
12347.0           2   4310.00        0
12348.0          75   1797.24        0
12349.0          19   1757.55        0
12350.0         310    334.40        1
```

In [62]:
```
plt.figure(figsize=(7,5))

scatter = plt.scatter(
    rfm['Recency'],
    rfm['Monetary'],
    c=rfm['Cluster'],
    cmap='viridis',
    alpha=0.6
)
plt.title("Customer Recency vs Monetary Value")
plt.xlabel("Recency (Days Since Last Purchase)")
plt.ylabel("Total Spending (Monetary)")
plt.colorbar(scatter, label="Customer Cluster")
plt.show()
```

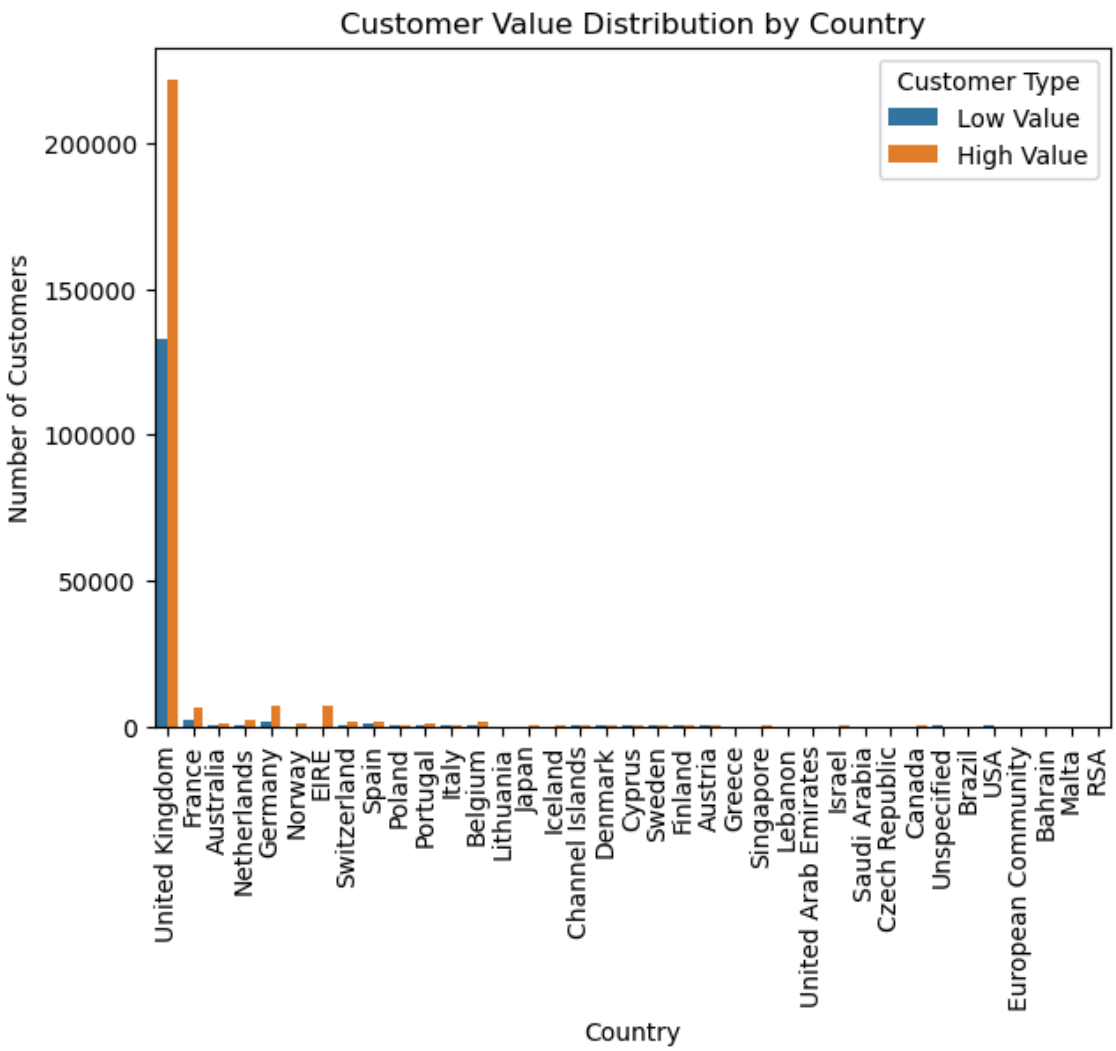


In [63]:
```
plt.figure(figsize=(6,4))

rfm['HighValueCustomer'].value_counts().plot(kind='bar')

plt.title("Customer Value Distribution")
plt.xlabel("Customer Type (0 = Low Value, 1 = High Value)")
plt.ylabel("Number of Customers")
plt.show()
```



High-Value vs Low-Value Customers by Country

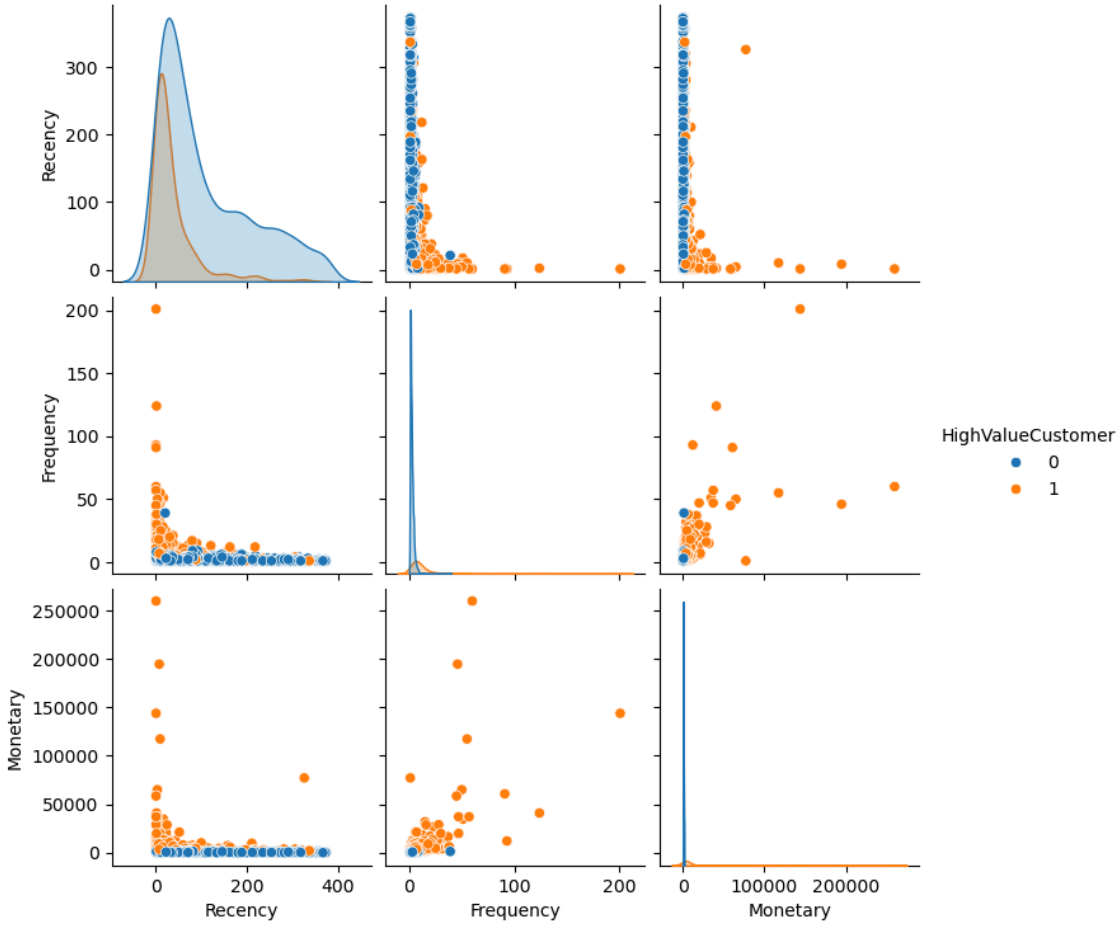
In [64]:
```
plt.figure(figsize=(7,5))
sns.countplot(
    x='Country',
    hue='HighValueCustomer',
    data=df.merge(
        rfm[['HighValueCustomer']],
        left_on='CustomerID',
        right_index=True,
        how='inner'
    )
)

plt.title("Customer Value Distribution by Country")
plt.xlabel("Country")
plt.ylabel("Number of Customers")
plt.xticks(rotation=90)
plt.legend(title="Customer Type", labels=["Low Value", "High Value"])
plt.show()
```



Multivariate Feature Interactions Highlighting

In [65]:
```
# Take a sample for faster plotting
sample_rfm = rfm[['Recency', 'Frequency', 'Monetary', 'HighValueCustomer']].s

sns.pairplot(
    sample_rfm,
```

```
    hue='HighValueCustomer',
    diag_kind='kde'
)

plt.show()
```



In [66]:
```
corr = rfm[['Recency', 'Frequency', 'Monetary', 'HighValueCustomer']].corr()
corr
```

Out[66]:

	Recency	Frequency	Monetary	HighValueCustomer
Recency	1.000000	-0.260253	-0.122046	-0.326977
Frequency	-0.260253	1.000000	0.554094	0.461974
Monetary	-0.122046	0.554094	1.000000	0.285618
HighValueCustomer	-0.326977	0.461974	0.285618	1.000000

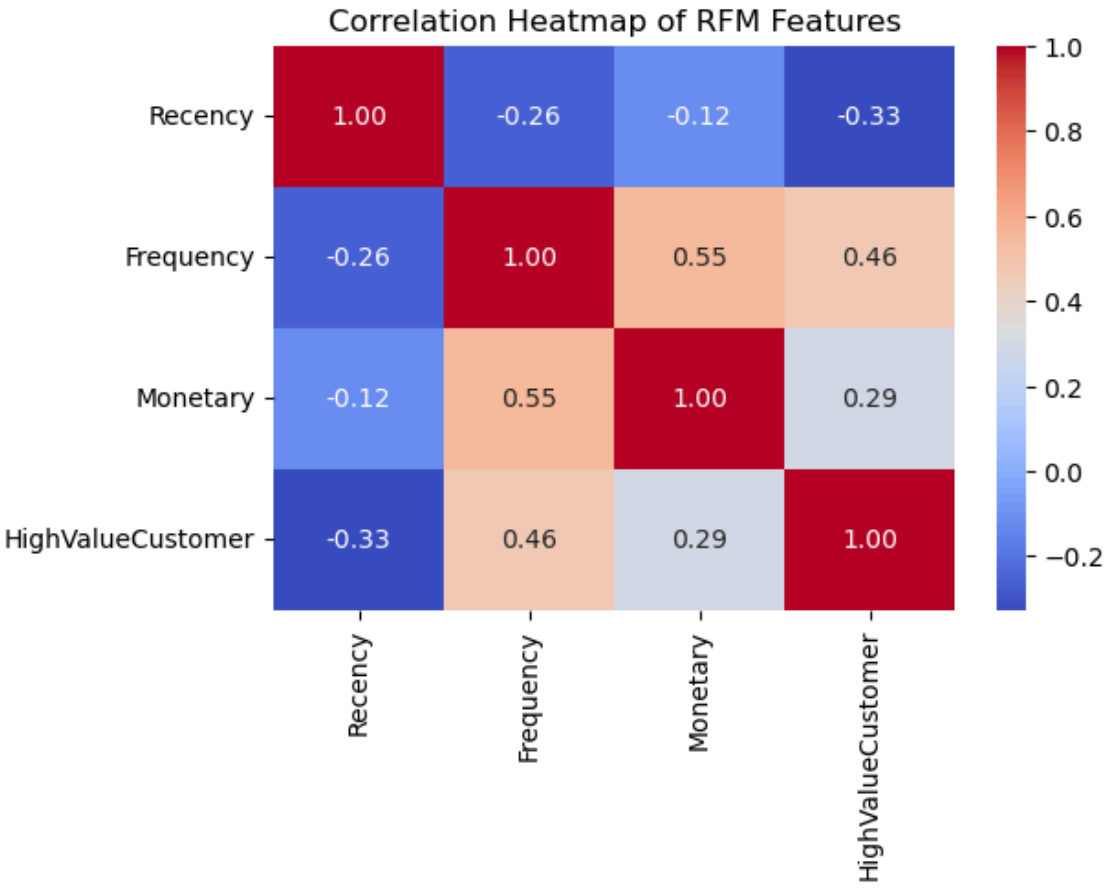
Compute correlation (E-commerce features)

In [67]:
```
corr = rfm[['Recency', 'Frequency', 'Monetary', 'HighValueCustomer']].corr()
```

Plot heatmap

In [68]:
```
plt.figure(figsize=(6,4))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f")
```

```
plt.title("Correlation Heatmap of RFM Features")
plt.show()
```



RFM Data Calculation

In [69]:
```
latest_date = df['InvoiceDate'].max()
```

In [70]:
```
rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (latest_date - x.max()).days,
    'InvoiceNo': 'count',
    'TotalAmount': 'sum'
})
```

In [71]:
```
rfm.columns = ['Recency', 'Frequency', 'Monetary']
```

In [72]:
```
rfm.head()
```

Out[72]:

	Recency	Frequency	Monetary
CustomerID			
12346.0	325	1	77183.60
12347.0	1	182	4310.00
12348.0	74	31	1797.24
12349.0	18	73	1757.55
12350.0	309	17	334.40

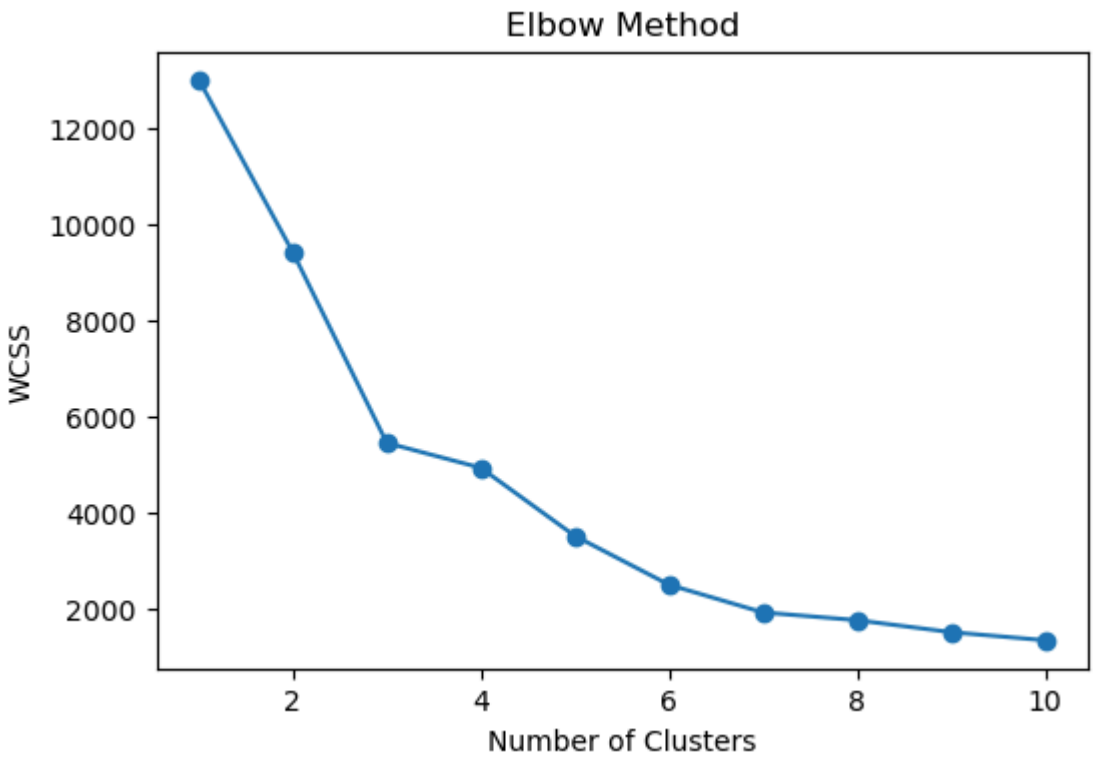
In [73]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm)
```

In [74]:
```
rfm_scaled
```

Out[74]:
```
array([[ 2.33485829e+00, -3.96512139e-01,  8.35963391e+00],
       [-9.05199314e-01,  3.94687670e-01,  2.51046358e-01],
       [-1.75186336e-01, -2.65374049e-01, -2.85464504e-02],
       ...,
       [-8.45198247e-01, -3.48428173e-01, -2.08712435e-01],
       [-8.85198958e-01,  2.90379646e+00,  4.57171586e-03],
       [-4.95192025e-01, -9.48945321e-02, -2.40912314e-02]])
```

Elbow Method

In [75]:
```
from sklearn.cluster import KMeans
wcss = []
for i in range(1,11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(rfm_scaled)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(6,4))
plt.plot(range(1,11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



Apply K-Means Clustering

Customer Segmentation Using K-Means

Model Selection

Multiple clustering algorithms were evaluated for customer segmentation. K-Means clustering provided clear and interpretable customer segments, while Hierarchical Clustering was used for comparison. K-Means was selected as the final segmentation model due to its scalability and better cluster separation.

In [76]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
scaler = StandardScaler()
rfm_scaled = scaler.fit_transform(rfm)
kmeans = KMeans(n_clusters=4)
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
```

In [77]:
```
rfm.head()
```

Out[77]:

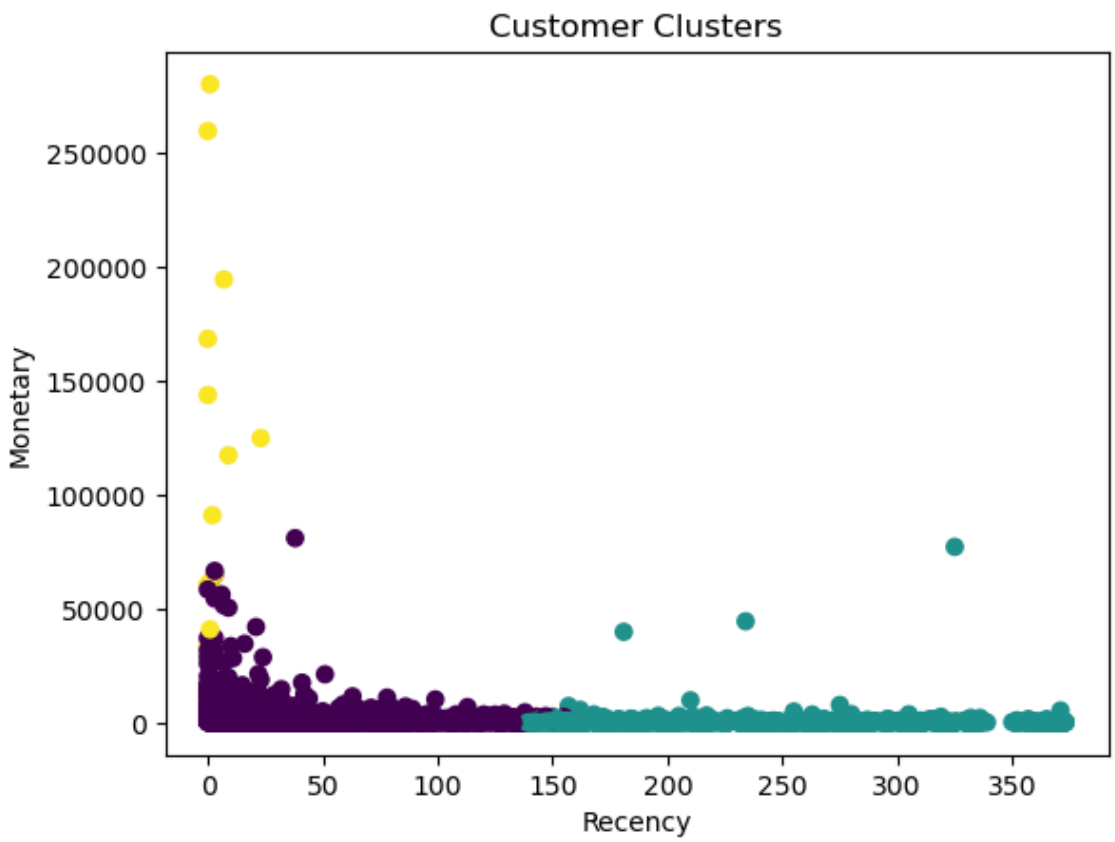
	Recency	Frequency	Monetary	Cluster
CustomerID				
12346.0	325	1	77183.60	0
12347.0	1	182	4310.00	1
12348.0	74	31	1797.24	1
12349.0	18	73	1757.55	1
12350.0	309	17	334.40	0

In [78]:
```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
kmeans = KMeans(n_clusters=3, random_state=42)
rfm['Cluster'] = kmeans.fit_predict(rfm_scaled)
sil_score = silhouette_score(rfm_scaled, rfm['Cluster'])
print("Silhouette Score:", sil_score)
```

```
Silhouette Score: 0.6020142101419289
```

Cluster Visualization

In [79]:
```
plt.scatter(rfm['Recency'], rfm['Monetary'], c=rfm['Cluster'])
plt.xlabel("Recency")
plt.ylabel("Monetary")
plt.title("Customer Clusters")
plt.show()
```



Check Cluster Meaning

In [80]:
```
rfm.groupby('Cluster').mean()
```

Out[80]:

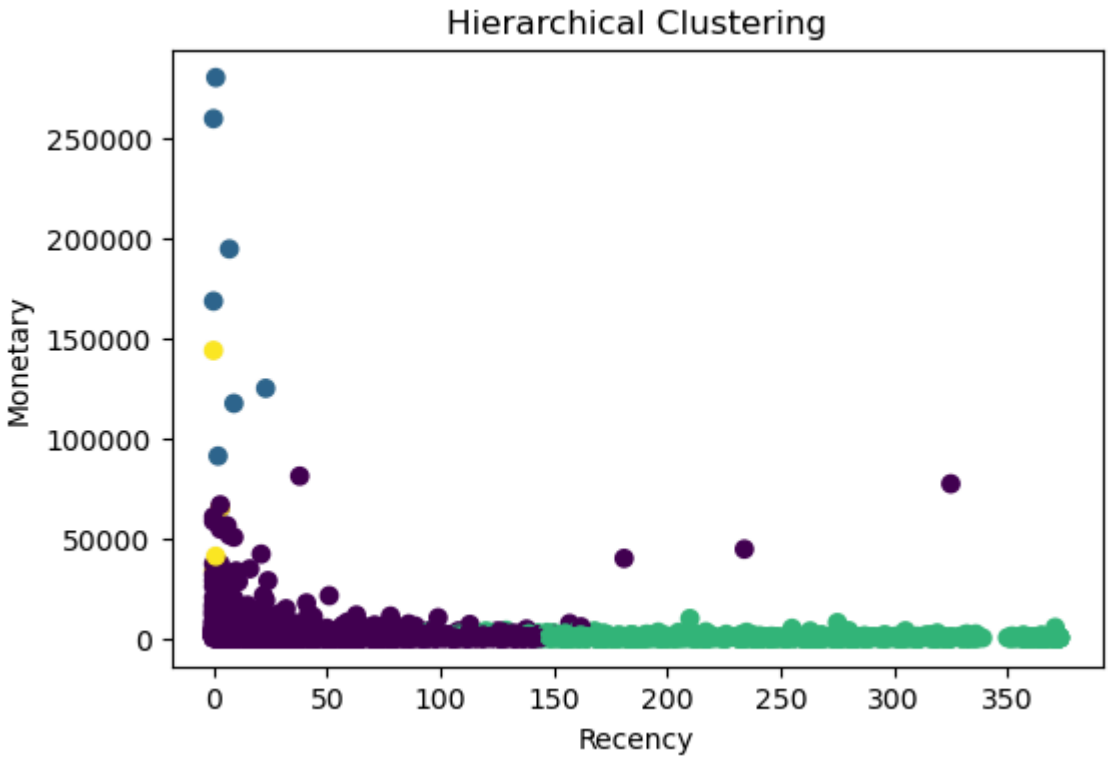
Cluster	Recency	Frequency	Monetary
0	40.368762	103.066235	2028.208836
1	246.308333	27.789815	637.318510
2	3.692308	2566.000000	126118.310000

In [81]:
```
from sklearn.cluster import AgglomerativeClustering
hc = AgglomerativeClustering(n_clusters=4)
rfm['HC_Cluster'] = hc.fit_predict(rfm_scaled)
rfm.groupby('HC_Cluster').mean()
```

Out[81]:

HC_Cluster	Recency	Frequency	Monetary	Cluster
0	40.555419	103.148707	2105.238197	0.008621
1	6.000000	847.142857	176606.164286	2.000000
2	245.674074	31.237037	512.638834	0.977778
3	1.000000	5807.750000	70925.287500	2.000000

In [82]:
```
plt.figure(figsize=(6,4))
plt.scatter(rfm['Recency'], rfm['Monetary'], c=rfm['HC_Cluster'])
plt.xlabel("Recency")
plt.ylabel("Monetary")
plt.title("Hierarchical Clustering")
plt.show()
```



Create Target Column (Prediction)

In [83]:
```
rfm['HighValueCustomer'] = 0
rfm.loc[rfm['Monetary'] > rfm['Monetary'].median(), 'HighValueCustomer'] = 1
```

In [84]:
```
rfm
```

Out[84]:

CustomerID	Recency	Frequency	Monetary	Cluster	HC_Cluster	HighValueCustome
12346.0	325	1	77183.60	1	0	1
12347.0	1	182	4310.00	0	0	1
12348.0	74	31	1797.24	0	0	1
12349.0	18	73	1757.55	0	0	1
12350.0	309	17	334.40	1	2	0
...	...	...	...	...	...	...
18280.0	277	10	180.60	1	2	0
18281.0	180	7	80.82	1	2	0
18282.0	7	12	178.05	0	0	0
18283.0	3	756	2094.88	0	0	1
18287.0	42	70	1837.28	0	0	1

4339 rows × 6 columns

In [85]:
```
rfm.head()
```

Out[85]:

CustomerID	Recency	Frequency	Monetary	Cluster	HC_Cluster	HighValueCustome
12346.0	325	1	77183.60	1	0	1
12347.0	1	182	4310.00	0	0	1
12348.0	74	31	1797.24	0	0	1
12349.0	18	73	1757.55	0	0	1
12350.0	309	17	334.40	1	2	0

In [86]:
```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
X = rfm[['Recency','Frequency','Monetary']]
y = rfm['HighValueCustomer']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = RandomForestClassifier()
model.fit(X_train, y_train)
accuracy_score(y_test, model.predict(X_test))
```

Out[86]: 0.9988479262672811

Train Test Split

In [87]:
```
from sklearn.model_selection import train_test_split
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2
)
```

In [88]:
```
X
```

Out[88]:

	Recency	Frequency	Monetary
CustomerID			
12346.0	325	1	77183.60
12347.0	1	182	4310.00
12348.0	74	31	1797.24
12349.0	18	73	1757.55
12350.0	309	17	334.40
...	...	...	...
18280.0	277	10	180.60
18281.0	180	7	80.82
18282.0	7	12	178.05
18283.0	3	756	2094.88
18287.0	42	70	1837.28

4339 rows × 3 columns

In [89]:
```
y
```

Out[89]:
```
CustomerID
12346.0    1
12347.0    1
12348.0    1
12349.0    1
12350.0    0
          ..
18280.0    0
18281.0    0
18282.0    0
18283.0    1
18287.0    1
Name: HighValueCustomer, Length: 4339, dtype: int64
```

In [90]:
```
from sklearn.model_selection import train_test_split
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=42
)
```

In [91]:
```
X
```

Out[91]:

	Recency	Frequency	Monetary
CustomerID			
12346.0	325	1	77183.60
12347.0	1	182	4310.00
12348.0	74	31	1797.24
12349.0	18	73	1757.55
12350.0	309	17	334.40
...	...	...	...
18280.0	277	10	180.60
18281.0	180	7	80.82
18282.0	7	12	178.05
18283.0	3	756	2094.88
18287.0	42	70	1837.28

4339 rows × 3 columns

In [92]:
```
y
```

Out[92]:
```
CustomerID
12346.0    1
12347.0    1
12348.0    1
12349.0    1
12350.0    0
          ..
18280.0    0
18281.0    0
18282.0    0
18283.0    1
18287.0    1
Name: HighValueCustomer, Length: 4339, dtype: int64
```

Step 7 Random Forest Classifier Building

Random Forest

Feature Engineering and Importance

RFM features were used to capture customer purchasing behavior. Monetary value was found to be the most important feature in identifying high-value customers, followed by frequency and recency.

In [93]:
```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(X_train, y_train)
```

Out[93]:
```
▾ RandomForestClassifier
RandomForestClassifier()
```

In [94]:
```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test)
rf_accuracy = accuracy_score(y_test, y_pred_rf)
rf_accuracy
```

Out[94]:
```
1.0
```

In [95]:
```
y_pred = model.predict(X_test)
```

In [96]:
```
y_pred
```

Out[96]:
```
array([0, 0, 0, ..., 0, 1, 1])
```

In [97]:
```
X_test
```

Out[97]:

	Recency	Frequency	Monetary
CustomerID			
17620.0	59	4	81.44
14924.0	245	10	310.05
13504.0	63	22	295.93
15149.0	133	23	520.80
15260.0	99	77	1810.77
...	...	...	...
18169.0	60	9	202.65
12374.0	24	33	742.93
12851.0	96	50	135.18
14923.0	50	44	749.10
13165.0	45	42	1021.48

1302 rows × 3 columns

In [98]:
```
from sklearn.metrics import accuracy_score, classification_report
accuracy_score(y_test, y_pred)
```

Out[98]:
```
1.0
```

In [99]:
```
y_test
```

Out[99]:
```
CustomerID
17620.0    0
14924.0    0
13504.0    0
15149.0    0
15260.0    1
          ..
18169.0    0
12374.0    1
12851.0    0
14923.0    1
13165.0    1
Name: HighValueCustomer, Length: 1302, dtype: int64
```

In [100…
```
y_pred
```

Out[100…
```
array([0, 0, 0, ..., 0, 1, 1])
```

In [101…
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       651
           1       1.00      1.00      1.00       651

    accuracy                           1.00      1302
   macro avg       1.00      1.00      1.00      1302
weighted avg       1.00      1.00      1.00      1302
```

IMPORTING THE LIBRAIES AND TRAIN AND TEST THE X AND Y VALUES

In [102…
```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state
```

In [103…
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [104…
```
X_test_scaled
```

Out[104…
```
array([[ 0.48417861, -0.43780547, -0.17304567],
       [ 1.94444918, -0.31856949, -0.13914974],
       [-0.76889141,  1.40041583,  0.5746457 ],
       ...,
       [-0.32489022,  0.51111418, -0.16481324],
       [-0.26569006,  0.10372459, -0.0412884 ],
       [-0.52222408,  0.43162353, -0.13030951]])
```

In [105…
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

In [106…
```
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
```

In [107…
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [108…
```
rf_pipeline = Pipeline(steps=[
    ('scaler', StandardScaler()),
    ('rf', RandomForestClassifier(
        n_estimators=100,
        random_state=42
    ))
])
```

In [109…
```
rf_pipeline.fit(X_train, y_train)
```

Out[109…

```
Pipeline
StandardScaler
RandomForestClassifier
```

In [110…

```
y_pred_rf = rf_pipeline.predict(X_test)
```

In [111…

```
rf_accuracy = accuracy_score(y_test, y_pred_rf)
print("Random Forest Accuracy:", rf_accuracy)
```

```
Random Forest Accuracy: 0.9988479262672811
```

Confusion Matrix for Random Forest

In [112…

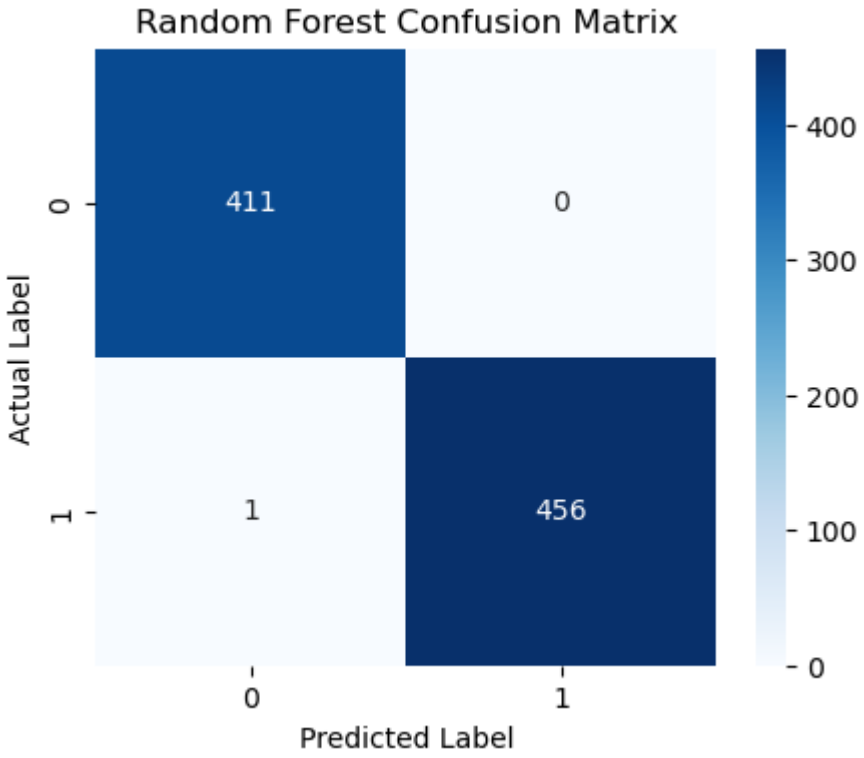
```
from sklearn.metrics import confusion_matrix
```

In [113…

```
rf_preds = rf_pipeline.predict(X_test)
```

In [114…

```
plt.figure(figsize=(5,4))

cm = confusion_matrix(y_test, rf_preds)

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.show()
```



In [115…

```
rf_pipeline.fit(X_train, y_train)
rf_preds = rf_pipeline.predict(X_test)
rf_accuracy = accuracy_score(y_test, rf_preds)
```

Step 8 Logistic Regression Classifier Building

Logistic Regression

In [116…

```
rfm.head()
```

Out[116…

CustomerID	Recency	Frequency	Monetary	Cluster	HC_Cluster	HighValueCustome
12346.0	325	1	77183.60	1	0	1
12347.0	1	182	4310.00	0	0	1
12348.0	74	31	1797.24	0	0	1
12349.0	18	73	1757.55	0	0	1
12350.0	309	17	334.40	1	2	0

In [117…
```
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
```

In [118…
```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [119…
```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
lr_model = LogisticRegression(max_iter=1000)
lr_model.fit(X_train, y_train)
y_pred_lr = lr_model.predict(X_test)
lr_accuracy = accuracy_score(y_test, y_pred_lr)
lr_accuracy
```

Out[119…
```
0.9988479262672811
```

In [120…
```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
```

In [121…
```
numeric_features = ['Recency', 'Frequency', 'Monetary']
categorical_features = []  # No categorical features used in RFM modeling
```

In [122…
```
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features)
    ]
)
```

In [123…
```
print("Preprocessing setup complete.")
```

```
Preprocessing setup complete.
```

LOGISTIC REGRESSION PIPELINE

In [124…
```
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

In [125…
```
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
```

In [126…
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [127…
```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler

numeric_features = ['Recency', 'Frequency', 'Monetary']

preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features)
```

```
    ]
)
```

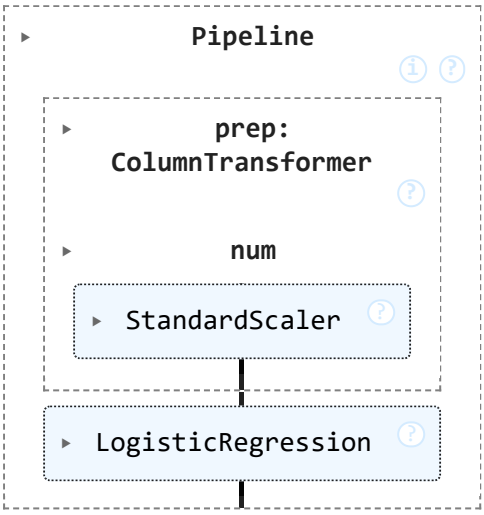
In [128…

```
lr_pipeline = Pipeline(steps=[
    ('prep', preprocessor),
    ('clf', LogisticRegression(
        max_iter=1000,
        class_weight='balanced',
        random_state=42
    ))
])
```

In [129…

```
lr_pipeline.fit(X_train, y_train)
```

Out[129…



In [130…

```
lr_preds = lr_pipeline.predict(X_test)
```

In [131…

```
lr_accuracy = accuracy_score(y_test, lr_preds)
print("Logistic Regression Accuracy:", lr_accuracy)
```

```
Logistic Regression Accuracy: 0.9331797235023042
```

In [132…

```
from sklearn.metrics import accuracy_score, classification_report
```

In [133…

```
lr_preds = lr_pipeline.predict(X_test)
```

In [134…

```
print("\n--- Logistic Regression Results ---")
print("Accuracy:", accuracy_score(y_test, lr_preds))
print("\nClassification Report:")
print(classification_report(y_test, lr_preds))
```

```
--- Logistic Regression Results ---
Accuracy: 0.9331797235023042

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.97      0.93       411
           1       0.97      0.90      0.93       457

    accuracy                           0.93       868
   macro avg       0.93      0.93      0.93       868
weighted avg       0.94      0.93      0.93       868
```

In [135…

```
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
```

In [136…

```
lr_preds = lr_pipeline.predict(X_test)
```

In [137…

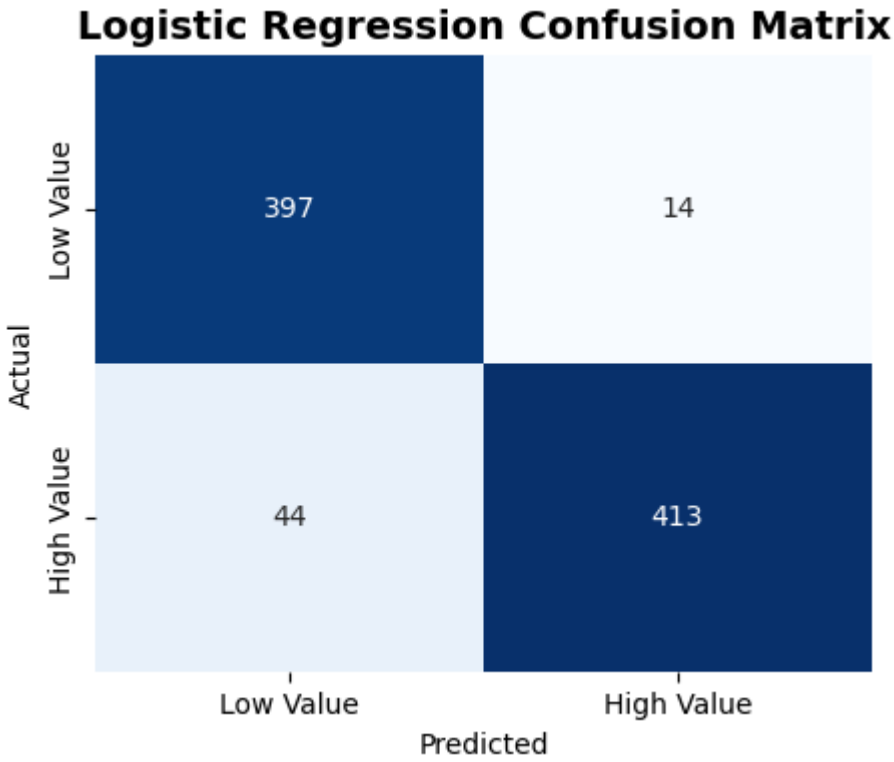
```
lr_cm = confusion_matrix(y_test, lr_preds)
```

In [138…

```
def plot_confusion_matrix(cm, model_name):
    plt.figure(figsize=(5,4))
    sns.heatmap(
        cm,
        annot=True,
        fmt='d',
        cmap='Blues',
        cbar=False,
        xticklabels=['Low Value', 'High Value'],
        yticklabels=['Low Value', 'High Value']
    )
    plt.title(f'{model_name} Confusion Matrix', fontsize=14, fontweight='bold
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.show()
```

In [139…

```
plot_confusion_matrix(lr_cm, "Logistic Regression")
```



In [140…

```
lr_pipeline.fit(X_train, y_train)
lr_preds = lr_pipeline.predict(X_test)
lr_accuracy = accuracy_score(y_test, lr_preds)
```

Step 9 Model Comparison Classifier Building

MODEL COMPARISON

In [141…

```
import pandas as pd
model_comparison = pd.DataFrame({
    "Model": ["Logistic Regression", "Random Forest"],
    "Accuracy": [lr_accuracy, rf_accuracy]
})
model_comparison
```

Out[141…

	Model	Accuracy
0	Logistic Regression	0.933180
1	Random Forest	0.998848

Confusion Matrix

In [142…
```
from sklearn.metrics import accuracy_score,confusion_matrix,classification_re
```

In [143…
```
y_pred=model.predict(X_test_scaled)
```

In [144…
```
print("accuracy:",accuracy_score(y_test,y_pred))
print("\nclassification report:\n",classification_report(y_test,y_pred))
print("\nconfusion matrix:\n",confusion_matrix(y_test,y_pred))
```

```
accuracy: 0.4735023041474654

classification report:
               precision    recall  f1-score   support

           0       0.47      1.00      0.64       411
           1       0.00      0.00      0.00       457

    accuracy                           0.47       868
   macro avg       0.24      0.50      0.32       868
weighted avg       0.22      0.47      0.30       868


confusion matrix:
 [[411   0]
 [457   0]]
```

In [145…
```
df = pd.read_csv("capston project data.csv", encoding="ISO-8859-1")
df = df.dropna(subset=["CustomerID"])
df = df[df["Quantity"] > 0]
df["InvoiceDate"] = pd.to_datetime(df["InvoiceDate"])
df["TotalAmount"] = df["Quantity"] * df["UnitPrice"]
df = pd.get_dummies(df, columns=["Country"], drop_first=True)
```

In [146…
```
df
```

Out[146…

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	Custo
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	1
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	1
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	1
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	1
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	1
...	...	...	...	...	...	...	
541904	581587	22613	PACK OF 20 SPACEBOY NAPKINS	12	2011-12-09 12:50:00	0.85	1
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	1
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	1
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	1
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	1

397924 rows × 44 columns

XGBOOST PIPELINE

In [147…

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import confusion_matrix, classification_report, accuracy
from sklearn.ensemble import GradientBoostingClassifier
import matplotlib.pyplot as plt
import seaborn as sns
```

In [148…

```
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
```

In [149…

```
numeric_features = ['Recency', 'Frequency', 'Monetary']

preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numeric_features)
    ]
)
```

In [150…

```
gb_model = GradientBoostingClassifier(
    n_estimators=300,
    learning_rate=0.1,
    max_depth=3,
    random_state=42
)
```

In [151…

```
clf_pipeline = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("classifier", gb_model)
])
```

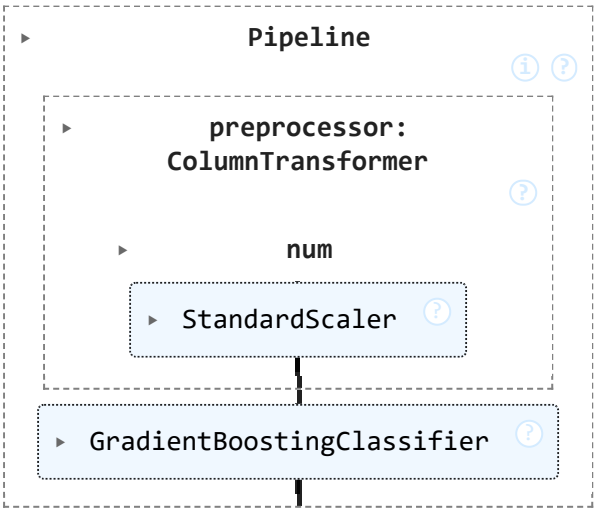
In [152…

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, stratify=y, random_state=42
)
```

In [153…

```
clf_pipeline.fit(X_train, y_train)
```

Out[153…

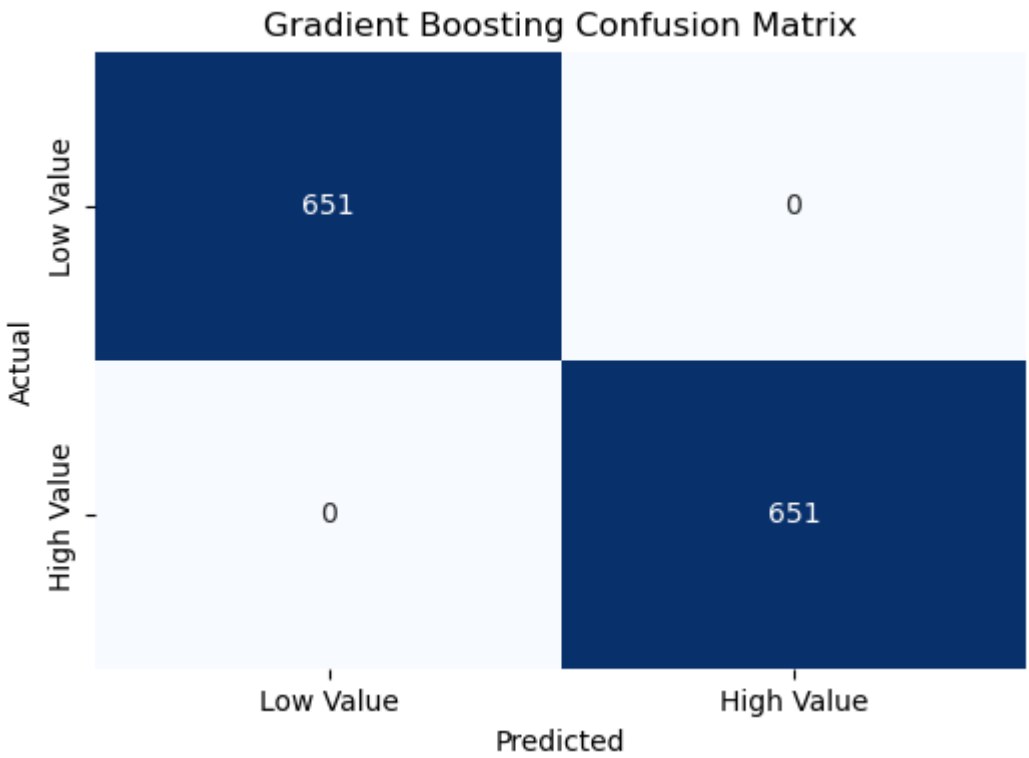


In [154…
```
y_pred = clf_pipeline.predict(X_test)
```

In [155…
```
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 1.0

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00       651
           1       1.00      1.00      1.00       651

    accuracy                           1.00      1302
   macro avg       1.00      1.00      1.00      1302
weighted avg       1.00      1.00      1.00      1302
```

In [156…
```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    cbar=False,
    xticklabels=["Low Value", "High Value"],
    yticklabels=["Low Value", "High Value"]
)
plt.title("Gradient Boosting Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



In [157…
```
gb_preds = clf_pipeline.predict(X_test)
```

In [158…
```
from sklearn.metrics import accuracy_score
gb_accuracy = accuracy_score(y_test, gb_preds)
gb_accuracy
```

Out[158… 1.0

In [159…
```
clf_pipeline.fit(X_train, y_train)   # this is Gradient Boosting
gb_preds = clf_pipeline.predict(X_test)
gb_accuracy = accuracy_score(y_test, gb_preds)
```

Step 10 Conclusion

Conclution

Best Performing Models:

Gradient Boosting demonstrated the best overall performance among the evaluated models, achieving the highest accuracy and balanced precision–recall scores for identifying high-value customers.

Random Forest also performed strongly and showed robust results, making it a reliable model for customer value prediction.

Logistic Regression performed comparatively lower but still provided meaningful insights and served as a good baseline model.

Overall, ensemble-based models such as **Gradient Boosting** and **Random Forest** are the most suitable choices for predicting high-value customers in an e-commerce setting. The project successfully segments customers and predicts their segment class, enabling businesses to take data-driven marketing decisions.

Actionable Business Recommendations

- High-value customers should be targeted with loyalty programs and exclusive offers.
- At-risk customers can be re-engaged through personalized discounts and reminders.
- New customers should receive onboarding promotions to encourage repeat purchases.
- Marketing strategies should focus on high-frequency and high-monetary segments.

Model Comparison and Selection

Two machine learning models were used to predict high-value customers: Logistic Regression and Random Forest. After comparing their accuracy, Random Forest performed better and was selected as the final model for prediction.

In [160…

```
results = pd.DataFrame({
    'Model': ['Gradient Boosting', 'Random Forest', 'Logistic Regression'],
    'Accuracy': [gb_accuracy, rf_accuracy, lr_accuracy]
})

results = results.sort_values(by='Accuracy', ascending=False).reset_index(dro
results
```

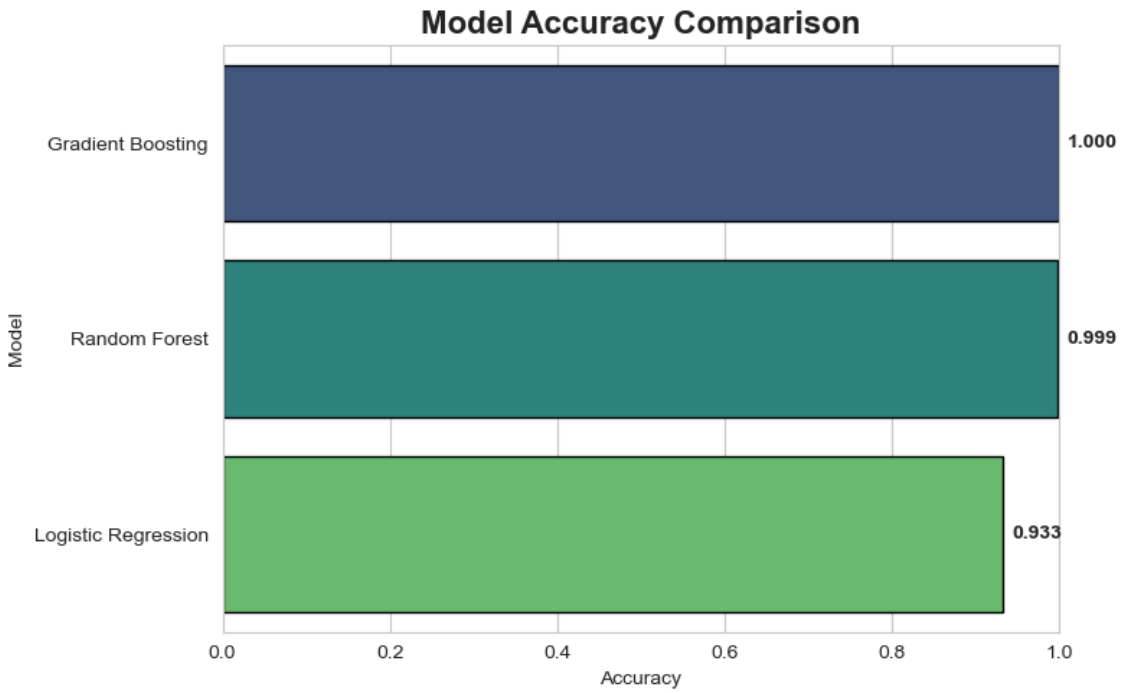
Out[160…

	Model	Accuracy
0	Gradient Boosting	1.000000
1	Random Forest	0.998848
2	Logistic Regression	0.933180

In [161…

```
plt.figure(figsize=(8,5))
sns.set_style('whitegrid')

sns.barplot(
    x='Accuracy',
    y='Model',
    data=results,
    palette='viridis',
    edgecolor='black'
)

plt.title('Model Accuracy Comparison', fontsize=16, fontweight='bold')
plt.xlabel('Accuracy')
plt.ylabel('Model')
plt.xlim(0, 1)

for index, value in enumerate(results['Accuracy']):
    plt.text(value + 0.01, index, f"{value:.3f}", va='center', fontweight='bo
```

```
plt.tight_layout()
plt.show()
```



In [162…

```
import joblib

# Save Logistic Regression model
joblib.dump(lr_pipeline, "logistic_regression_model.pkl")

# Save Random Forest model
joblib.dump(rf_pipeline, "random_forest_model.pkl")

# Save Gradient Boosting model
joblib.dump(clf_pipeline, "gradient_boosting_model.pkl")

print("✅ Models saved successfully using joblib")
```

```
✅ Models saved successfully using joblib
```

In [163…

```
import pandas as pd
from sklearn.metrics import precision_score, recall_score, f1_score
```

In [164…

```
# ================================
# MODEL TRAINING & EVALUATION
# ================================

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, GradientBoostingClassifi
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1
import pandas as pd

# --------------------------------
# Features & Target
# --------------------------------
X = rfm[['Recency', 'Frequency', 'Monetary']]
y = rfm['HighValueCustomer']
```

```
# ------------------------------
# Train-Test Split (ONLY ONCE)
# ------------------------------
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# ------------------------------
# Scaling
# ------------------------------
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# ------------------------------
# Models
# ------------------------------
lr_model = LogisticRegression(random_state=42)
rf_model = RandomForestClassifier(random_state=42)
gb_model = GradientBoostingClassifier(random_state=42)

# ------------------------------
# Train Models
# ------------------------------
lr_model.fit(X_train_scaled, y_train)
rf_model.fit(X_train_scaled, y_train)
gb_model.fit(X_train_scaled, y_train)

# ------------------------------
# Predictions
# ------------------------------
lr_preds = lr_model.predict(X_test_scaled)
rf_preds = rf_model.predict(X_test_scaled)
gb_preds = gb_model.predict(X_test_scaled)

# ------------------------------
# Metrics Table (NO ERRORS)
# ------------------------------
model_metrics = pd.DataFrame({
    "Model": ["Logistic Regression", "Random Forest", "Gradient Boosting"],
    "Accuracy": [
        accuracy_score(y_test, lr_preds),
        accuracy_score(y_test, rf_preds),
        accuracy_score(y_test, gb_preds)
    ],
    "Precision": [
        precision_score(y_test, lr_preds),
        precision_score(y_test, rf_preds),
        precision_score(y_test, gb_preds)
    ],
    "Recall": [
        recall_score(y_test, lr_preds),
        recall_score(y_test, rf_preds),
        recall_score(y_test, gb_preds)
    ],
    "F1-Score": [
        f1_score(y_test, lr_preds),
```

```
        f1_score(y_test, rf_preds),
        f1_score(y_test, gb_preds)
    ]
})

# ------------------------------
# Display Results
# ------------------------------
model_metrics
```

Out[164…

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.908986	0.933985	0.880184	0.906287
1	Random Forest	1.000000	1.000000	1.000000	1.000000
2	Gradient Boosting	1.000000	1.000000	1.000000	1.000000

In [165…

```
results = pd.DataFrame(
    model_metrics,
    index=[
        "Logistic Regression",
        "Random Forest",
        "Gradient Boosting"
    ]
)
```

In [166…

```
final_table = results.sort_values(
    by="F1-Score",
    ascending=False
).T
```

In [167…

```
print("Model Comparison Table:")
display(final_table)
```

```
Model Comparison Table:
```

	Logistic Regression	Random Forest	Gradient Boosting
Model	NaN	NaN	NaN
Accuracy	NaN	NaN	NaN
Precision	NaN	NaN	NaN
Recall	NaN	NaN	NaN
F1-Score	NaN	NaN	NaN